

CSE276 · ARTIFICIAL INTELLIGENCE FOUNDATIONS

UNIT 3

Intelligent Agents

What turns a program into something that perceives, decides and acts - and how five architectures answer that differently.

Agent Architecture • PEAS Framework • Rational Agents • Task Environments • Reflex • Model • Goal • Utility • Learning • Multi-Agent Systems • Human-AI Collaboration

UNIT III ROADMAP

What This Unit Covers

UNIT 3

One loop - perceive, decide, act - and five increasingly capable ways of filling in the middle step.

▪ Foundations

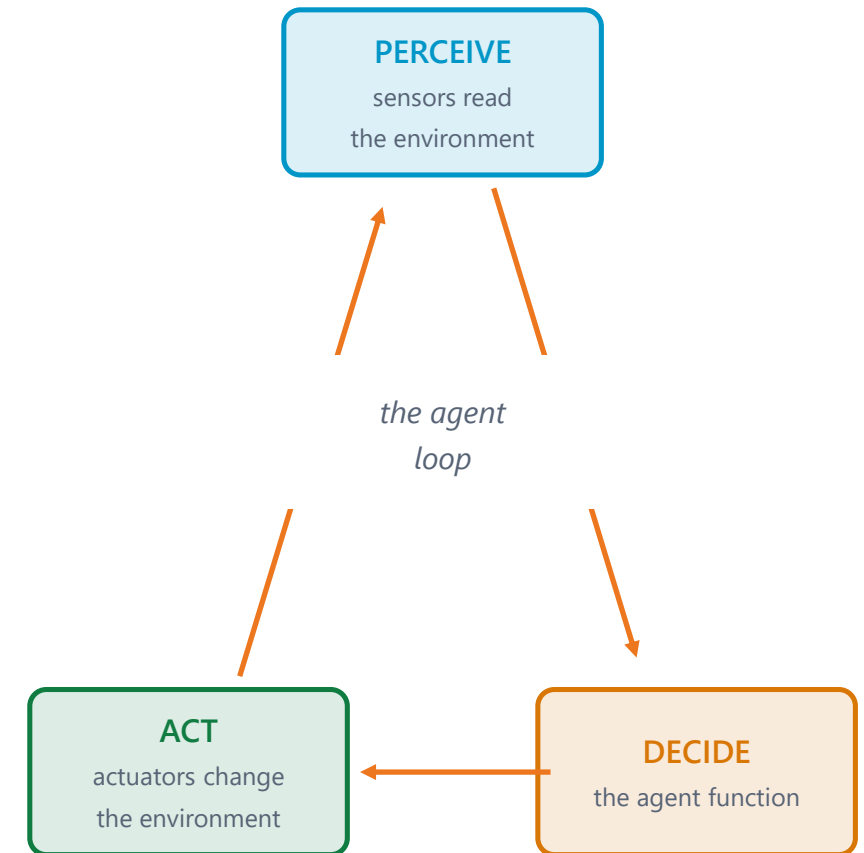
- Introduction to intelligent agents and agent architecture
- The PEAS framework: Performance, Environment, Actuators, Sensors
- Rational agents and the properties of task environments

▪ The five agent types

- Simple reflex agents and model-based agents
- Goal-based agents
- Utility-based agents and learning agents

▪ Beyond a single agent

- Autonomous intelligent systems and multi-agent systems
- Cooperation and coordination
- Human-AI collaboration and applications



01

SECTION

Agents and Architecture

IN THIS SECTION

- What an agent is, precisely
- The agent function and the agent program
- Why acting changes what you next perceive
- What separates an agent from an ordinary function

3.1 AGENTS AND ARCHITECTURE

What Exactly Is an Agent?

UNIT 3

KEY CONCEPT · The definition, and the property that makes it more than a function

The definition an agent perceives its environment through SENSORS and acts upon it through ACTUATORS.

The percept sequence everything the agent has perceived so far. Its behaviour may depend on all of it, not just the latest reading.

The agent function the abstract mapping from percept sequence to action. It describes WHAT the agent does.

The agent program the concrete implementation running on the hardware. It describes HOW.

THE KEY PROPERTY acting CHANGES the environment, and the changed environment is what the agent perceives next. That feedback is what separates an agent from a function.

A function takes an input, returns an output, and the world is unaffected. An agent acts, the world moves, and it must cope with the world it just changed.

TEST IT Is a calculator an agent? Is a thermostat?

3.1 AGENT ARCHITECTURE

Agent Architecture - The Two Halves

UNIT 3

The architecture sets the limits. The program decides what the agent does within them.

Architecture

What it is the physical or computational platform the agent runs on.

Provides the sensors, the actuators, and the compute that connects them.

Examples a robot chassis with cameras and motors; a server with API access; a phone.

Constrains what the agent can possibly perceive and possibly do.

The equation agent = architecture + program.

Agent program

What it is the decision logic that maps percepts to actions.

Ranges from a handful of IF-THEN rules to a learned policy.

This is the part you design and study in this unit.

Five standard forms simple reflex, model-based, goal-based, utility-based, learning.

Choice depends on the environment, not on which sounds most sophisticated.

02

SECTION

The PEAS Framework

IN THIS SECTION

- Performance measure, Environment, Actuators, Sensors
- Why you specify PEAS before designing anything
- Worked PEAS descriptions
- The mistake almost everyone makes with the performance measure

3.2 PEAS FRAMEWORK

UNIT 3

PEAS - Four Things to Specify Before You Design

Write these four down and most design decisions make themselves.

P

Performance measure

- How success is judged - by an OUTSIDE observer
- Must be measurable, not aspirational
- Specify the outcome you want, not the behaviour
- This is the one people get wrong

E

Environment

- Everything outside the agent that it interacts with
- Need not be physical - a simulator counts
- Its properties decide which architecture you need
- Characterised on the five axes in section 03

A

Actuators

- Everything the agent can use to ACT
- Motors, displays, API calls, messages
- Defines the set of possible actions
- No actuator means no agency

S

Sensors

- Everything the agent uses to PERCEIVE
- Cameras, lidar, keyboards, log files, web requests
- Defines what the agent can possibly know
- Missing sensors cause partial observability

3.2 PEAS PROFILES

PEAS Descriptions for Three Real Agents

UNIT 3

The same four headings, three completely different systems.

	Autonomous taxi	Medical diagnosis system	Warehouse robot
Performance	Safe, legal, fast, comfortable, low fuel	Correct diagnosis, low unnecessary testing, patient outcome	Orders fulfilled per hour, no collisions, low energy
Environment	Roads, traffic, pedestrians, weather, signage	Patient, symptoms, test results, medical history	Aisles, shelves, other robots, human staff
Actuators	Steering, accelerator, brake, indicators, horn	Questions, test requests, diagnosis and report	Wheels, lift mechanism, status lights
Sensors	Cameras, lidar, radar, GPS, speedometer	Patient answers, lab results, imaging	Cameras, bump sensors, wheel encoders, floor markers

3.2 PERFORMANCE MEASURE

Specify the Outcome, Not the Behaviour

UNIT 3

COMMON PITFALL · The classic performance-measure mistake

The example a vacuum-cleaning agent. You want a clean floor, so you reward the agent for the amount of dirt collected.

What the agent learns collect dirt, dump it back on the floor, collect it again. Its score is excellent and the floor is filthy.

What went wrong you rewarded the BEHAVIOUR you associated with success rather than success itself.

The fix reward the state of the world - a clean floor at each time step - not the activity.

Why this matters generally an agent optimises exactly what you wrote down, not what you meant. This is the same problem as reward design in reinforcement learning.

The habit before you write a performance measure, ask what an agent could do to score highly while defeating your intention.

TRY IT Reward a delivery robot for 'packages delivered per hour'. How could it cheat?

03

SECTION

Rational Agents and Task Environments

IN THIS SECTION

- What rationality actually requires
- Rationality is not omniscience
- Five properties of task environments
- Why the environment chooses the architecture

3.3 RATIONAL AGENTS

What a Rational Agent Is

UNIT 3

KEY CONCEPT · And four things it is not

The definition for each possible percept sequence, a rational agent selects the action expected to MAXIMIZE its performance measure, given the evidence and its built-in knowledge.

NOT omniscient omniscience means knowing the actual outcome. Rationality only requires the best decision on the information available.

NOT perfect a rational agent can act correctly and still get a bad result. Crossing a road after looking both ways is rational even if a cargo door falls on you.

NOT clairvoyant it cannot be expected to predict the unpredictable.

It IS expected to explore gathering information is itself an action, and failing to look before acting is irrational.

It IS expected to learn a rational agent should improve as its percept sequence grows, rather than repeating an avoidable mistake.

3.3 TASK ENVIRONMENTS

UNIT 3

Five Properties of Task Environments

Characterise the environment first. It tells you which architecture you need.

PROPERTY 1

Fully vs partially observable

- Can the agent see the complete state?
- Chess: fully. Driving: partially
- Partial observability forces internal state

PROPERTY 2

Deterministic vs stochastic

- Does an action always give the same result?
- 8-puzzle: deterministic. Driving: stochastic
- Stochastic forces plan-act-observe-replan

PROPERTY 3

Episodic vs sequential

- Does this action affect future ones?
- Defect spotting: episodic. Chess: sequential
- Sequential requires lookahead

PROPERTY 4

Static vs dynamic

- Does the world change while you deliberate?
- Crossword: static. Driving: dynamic
- Dynamic puts a deadline on every decision

PROPERTY 5

Discrete vs continuous

- Finite distinct states, or a continuum?
- Chess: discrete. Steering: continuous
- Continuous rules out enumerating states

ALSO

Single vs multi-agent

- Are there other agents with their own goals?
- Competitive, like chess, or cooperative
- Multi-agent needs coordination - section 05

3.3 TASK ENVIRONMENTS

Why Chess Was Solved Before Driving

UNIT 3

Every row explains part of the gap. The hard environment, not the hard reasoning, is the obstacle.

	Chess	Driving
Observability	Fully observable - the whole board is visible	Partially observable - occlusions, blind spots, intentions
Determinism	Deterministic - a move always has the same effect	Stochastic - tyres slip, other drivers surprise you
Dynamics	Static - the board waits for you to think	Dynamic - the world moves while you deliberate
State space	Discrete - a finite set of legal positions	Continuous - steering angle and speed are real numbers
Performance measure	Unambiguous - win, lose or draw	Contested - safe, legal, fast and comfortable can conflict
Consequence	Deep Blue won in 1997	Still not fully solved decades later

04

SECTION

The Five Agent Types

IN THIS SECTION

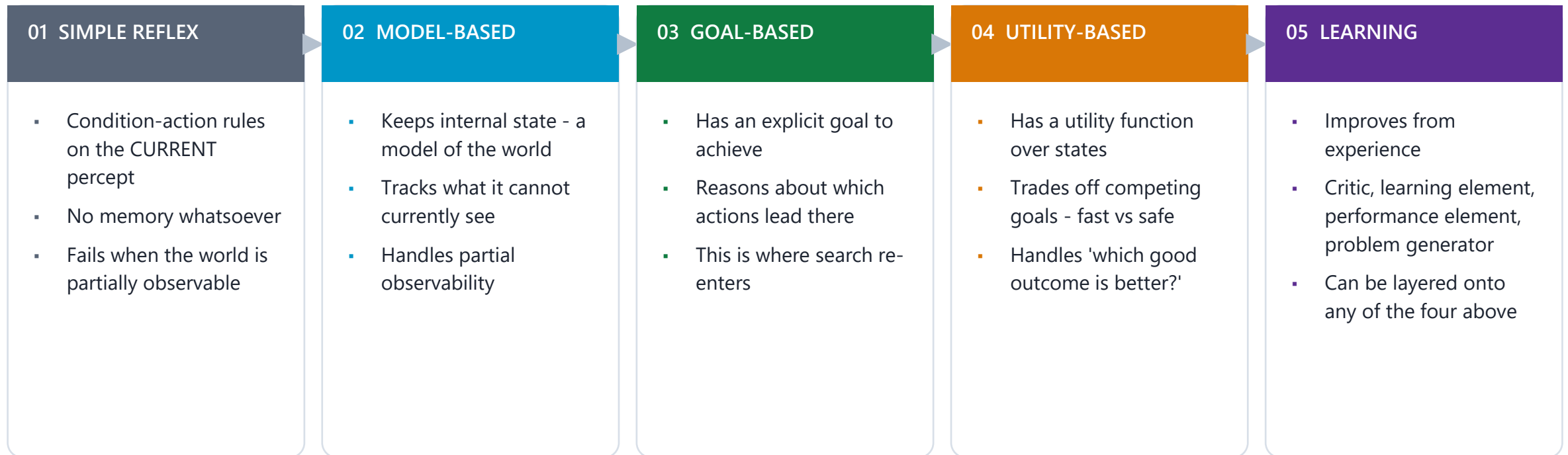
- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents and learning agents

3.4 TYPES OF AGENT

UNIT 3

Five Architectures of Increasing Capability

Each step adds one capability - and each is a response to a harder environment.



MATCH A robot vacuum that maps your house. Which type?

3.4 TYPES OF AGENT

What Each Agent Type Adds

UNIT 3

Read the last column: each type fails exactly where the next one's addition is needed.

Agent type	Adds	Handles	Fails when
Simple reflex	Nothing - rules only	Fully observable, simple worlds	The right action depends on unseen state
Model-based	Internal state	Partial observability	It has no goal, so it cannot plan
Goal-based	An explicit goal	Problems needing lookahead	Two goals conflict and it cannot choose
Utility-based	A utility function	Trade-offs between outcomes	The utility function is wrong or incomplete
Learning	Improvement from experience	Unknown environments	Feedback is absent, delayed or misleading

3.4 TYPES OF AGENT

More Sophisticated Is Not Automatically Better

UNIT 3

EXAM TIP · Match the architecture to the environment, not to ambition

The temptation to reach for the most capable architecture available, because it sounds better.

The reality in a fully observable, deterministic, static world, a simple reflex agent can be competitive AND far simpler to build, test and certify.

Every addition costs internal state must be maintained and can become stale. A goal needs a planner. A utility function needs someone to specify it correctly.

A wrong utility function is worse than no utility function - the agent will pursue it faithfully.

The design question which property of MY environment forces me up the ladder?

Answer that and the architecture chooses itself, with a justification you can defend.

CHECK YOUR UNDERSTANDING

Quick Check - Agents

UNIT 3

QUICK CHECK

An agent must clean a house it cannot see all of at once, and must remember which rooms it has already cleaned. What is the **SIMPLEST** architecture that suffices?

A Simple reflex agent

B Model-based reflex agent

C Utility-based agent

D Learning agent

ANSWER: B The requirement to remember what it has already done means the correct action depends on state the agent cannot currently perceive - so a simple reflex agent is insufficient. Internal state is exactly what a model-based agent adds. Nothing in the description requires goals, trade-offs or learning.

05

SECTION

Multi-Agent Systems and Human-AI Collaboration

IN THIS SECTION

- Autonomous intelligent systems
- Cooperation and coordination
- Why local optima and global optima diverge
- Human-AI collaboration and applications

3.5 MULTI-AGENT SYSTEMS

Single Agent vs Multi-Agent

UNIT 3

Adding agents does not just add work - it adds interaction.

What changes with more than one agent

The environment now contains other agents whose actions you cannot control and may not observe.

Cooperative agents share a goal - warehouse robots fulfilling orders, or a search-and-rescue team.

Competitive agents have opposing goals - chess, auctions, adversarial settings.

Mixed partly aligned - traffic, where everyone wants to arrive but the road is shared.

Communication becomes an action in its own right, with its own cost.

Coordination mechanisms

Central assignment one coordinator allocates tasks. Simple, optimal, and a single point of failure.

Auction or contract net agents bid for tasks and the lowest-cost bidder wins. Distributed and scalable.

Convention agreed rules, such as driving on one side, that remove the need to negotiate.

Negotiation agents exchange proposals until they agree.

The cost coordination consumes computation and communication, and it grows faster than the number of agents.

3.5 COOPERATION AND COORDINATION

UNIT 3

When Every Agent Is Optimal and the System Is Not

COMMON PITFALL · The central problem multi-agent systems exist to solve

The scenario several robots each choose the task nearest to themselves. Every one of them makes the individually optimal choice.

What happens two robots choose the SAME task, both travel there, and one arrives to find the work already done.

The result every agent acted optimally and the system performed worse than it needed to.

The name the gap between the local optimum and the global optimum.

The everyday example traffic. If everyone takes the route the app says is fastest, that route stops being fastest.

Why coordination exists not to make individual agents smarter, but to stop individually smart choices from colliding.

NAME ONE Where else have you seen individually rational choices produce a bad collective outcome?

3.6 HUMAN-AI COLLABORATION

Human-AI Collaboration and Applications

UNIT 3

Autonomy is a spectrum, not a switch - and the right point on it depends on the cost of being wrong.

MODEL 1

Human in the loop

- The agent proposes, a human approves
- Used where errors are costly or regulated
- Medical decision support, loan approval

MODEL 2

Human on the loop

- The agent acts; a human supervises and can intervene
- Used where the agent must be fast
- Autonomous vehicles, industrial control

MODEL 3

Agent as assistant

- The agent handles routine work; the human handles judgement
- Drafting, triage, scheduling, summarising
- The most common model in practice

APPLICATIONS

Where agents are deployed

- Warehouse and delivery robotics
- Traffic signal control and smart grids
- Disaster response coordination
- Personal and organisational assistants

UNIT III CONSOLIDATION

Activity: Specify an Agent

UNIT 3

GROUP WORK · 12 minutes

In groups of 3-4, fully specify one agent, then justify its architecture from the properties of its environment.

1

Choose an agent

- A campus delivery robot
- An adaptive traffic signal controller
- A hostel energy-management system
- Or propose your own

2

Write the PEAS description

- Performance measure - measurable, and outcome-based
- Environment, Actuators, Sensors
- Check: could an agent score well while defeating your intent?

3

Characterise the environment

- All five properties, with a one-line justification each
- Single or multi-agent?

4

Choose the architecture

- Which of the five types is the SIMPLEST that suffices?
- Name the environment property that forces you up the ladder

DELIVERABLE One speaker per group: 60 seconds giving your performance measure, your architecture, and the single environment property that decided it.

UNIT III RECAP

Unit III - The Seven Ideas Worth Remembering

UNIT 3

1

An agent perceives and acts, and acting changes what it perceives
That feedback loop is what separates an agent from an ordinary function.

3

PEAS specifies an agent before you build it
Performance measure, Environment, Actuators, Sensors. Write these four and most design decisions follow.

5

Rationality is not omniscience
A rational agent takes the best action given the evidence available. It can act correctly and still get a bad outcome.

7

Individually optimal agents can make a worse system
Coordination exists to stop individually smart choices from colliding.

2

Agent = architecture + program
The architecture provides sensors, actuators and compute; the program is the decision logic you design.

4

Specify the outcome, not the behaviour
An agent optimises exactly what you wrote down. Reward dirt collected and it will tip the bin out to collect it again.

6

The environment chooses the architecture
Partial observability forces internal state; conflicting goals force a utility function. Sophistication is a response, not a virtue.

NEXT UP Unit IV - Evolution of Modern AI: symbolic and statistical AI, the three types of machine learning, deep learning, and the foundation-model paradigm.